

Pooja Ravi

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EDUCATION

- MSc, Applied Computing** 2025
University of Toronto, Canada
Awards: Vector Scholarship in Artificial Intelligence (2023-24) | Mitacs Globalink Graduate Fellowship.
- BTech, Computer Science and Engineering** 2023
SRM Institute of Science and Technology, India

RESEARCH EXPERIENCE

Researcher | University of Toronto (TCIG) Apr 2026 - Present
Supervised by David B. Lindell

In-hand robotic object manipulation

- **Action recovery.** Tracked ray-traced hand-link, palm, and wrist query points with CoTracker; solved frame-wise inverse kinematics by minimizing reprojection error, warm-started from previous-frame angles. Refined solutions via constrained cross-entropy method (CEM) search within a radius of the IK angles to improve the intended manipulation motion.
- **Physics replay.** Executed recovered actions in MuJoCo with contact and gravity; passed a frame from each replay back as the next video-generation cycle's initial frame, closing the loop.
- **Closed-loop pipeline.** Integrated a fine-tuned object-motion-conditioned video model with CoTracker, inverse kinematics, and constrained CEM refinement for ORCA Hand v1, connecting generated video to refined joint-space actions and MuJoCo feedback.

ProxyPose

- **Benchmarking.** Engineered a unified 6-DoF benchmark across HO3D, YCBinEOAT, and YCB-Video; standardized meshes, RGB/depth frames, pose annotations, and metadata in a consistent evaluation format.
- **Evaluation.** Implemented task-aligned metrics and dataset-specific evaluation pipelines; integrated competing methods, ran cross-dataset experiments, and analyzed comparative results.

PUBLICATIONS

1. R. Zhang, F. Taubner, **P. Ravi**, K. N. Kutulakos, and D. B. Lindell. **ProxyPose: 6-DoF Pose Tracking via Video-to-Video Translation.** **NeurIPS** 2026
2. **P. Ravi**, S. Roy, I. Dutta, and K. Kottursamy. **Attention Mechanism, Linked Networks, and Pyramid Pooling Enabled 3D Biomedical Image Segmentation.** **IEEE/ACIS SNPD** 2022
3. **P. Ravi** and A. Dewan. **Real-time Multi-Module Student Engagement Detection System.** **ICCIS** 2022

APPLIED MACHINE LEARNING EXPERIENCE

Machine Learning Engineer | Q2 Software May 2024 - Mar 2026

- Led cheque-signature verification across 200K images, training variational autoencoder embeddings and Siamese networks with triplet and contrastive losses; fine-tuned multimodal LLMs for pairwise visual comparison.
- Trained LayoutLM, Donut (Swin+BART), and TrOCR on 100K+ cheques; improved text extraction by 8% and fraud flagging by 4% while reducing inference latency through model optimization.

Machine Learning Engineer | 16 Bit Oct 2023 - Mar 2024

- Led temporal X-ray image-to-image translation between baseline and follow-up scans (12-96 months) to study knee osteoarthritis progression; scaled GAN and diffusion experiments with PyTorch Lightning/LoRA on 70K+ images, achieving ~10x faster training and ~19% higher image realism.

SELECTED PROJECTS

Echo Charlie: multimodal voice restoration system; Boson AI Hackathon winner (2025).

2D Sketch to 3D Stylization: NeRF-based sketch stylization through image style transfer and CLIP-guided text prompts.

MobileGen: optimized DDPM for mobile inference using quantization-aware training, distillation, and pruning; 20% faster with 14% fewer GFLOPs.